SIEM SECURITY MONITORING PROJECT

Log Forwarding Architecture & Unauthorized Access Detection

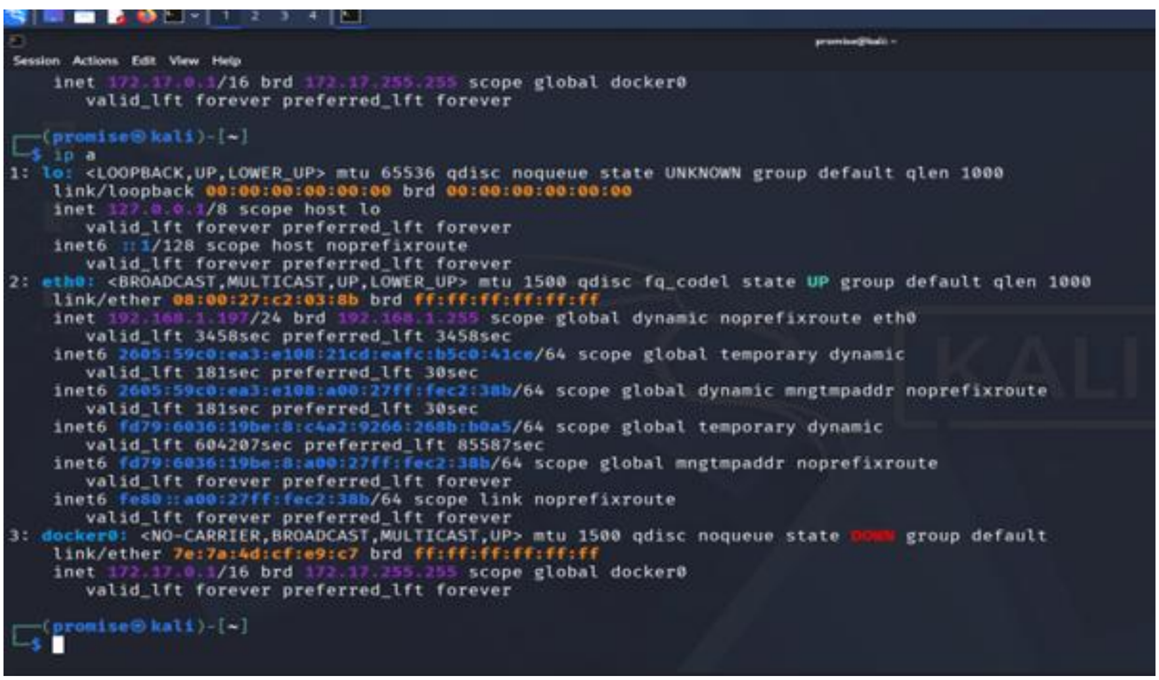
1. Project Overview

This project demonstrates the deployment of a centralized logging infrastructure in a hybrid environment. I configured a Kali Linux sensor to forward security logs to a Splunk Enterprise instance on Windows 11. This pipeline was validated by simulating and detecting unauthorized SSH authentication attempts in real-time.

2. Network Architecture

Component	OS	IP Address	Role
Sensor	Kali Linux	192.168.1.197	Traffic Generation & Log Source
Analyzer	Windows 11	192.168.1.229	Data Indexing & SIEM Visualization

EVIDENCE 01: SENSOR NETWORK CONFIGURATION

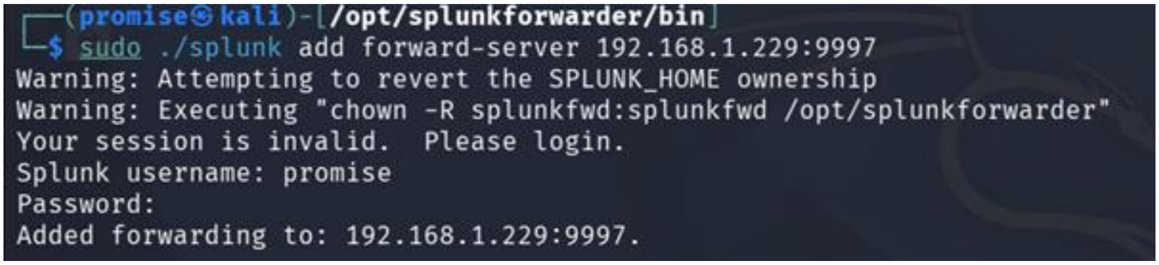


Verification of the network interface on the Kali Linux sensor.

3. Data Pipeline Implementation

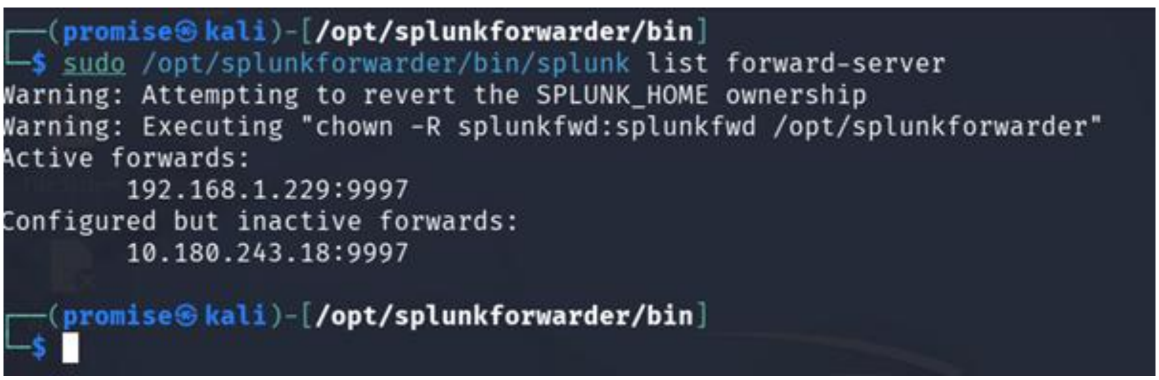
The Splunk Universal Forwarder was installed on the Kali sensor and configured to communicate with the receiver (Windows) via management port 9997.

EVIDENCE 02: DESTINATION CONFIGURATION



```
┌──(promise㉿kali)-[/opt/splunkforwarder/bin]
└─$ sudo ./splunk add forward-server 192.168.1.229:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid.  Please login.
Splunk username: promise
Password:
Added forwarding to: 192.168.1.229:9997.
```

EVIDENCE 03: HANDSHAKE VERIFICATION



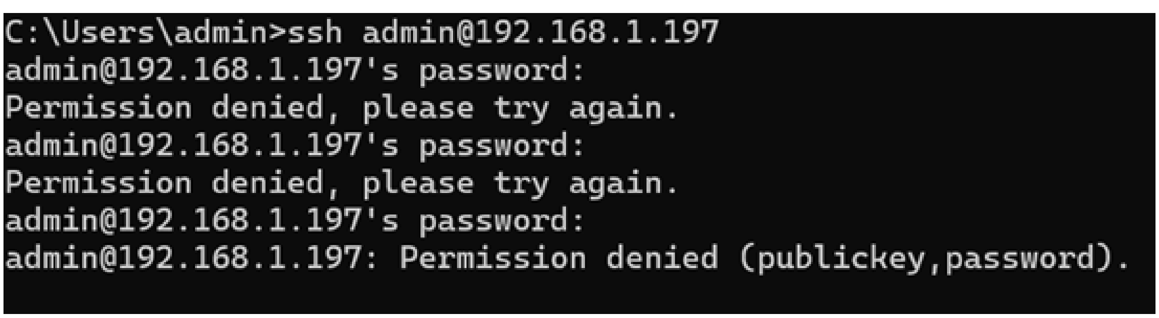
```
┌──(promise㉿kali)-[/opt/splunkforwarder/bin]
└─$ sudo /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Active forwards:
        192.168.1.229:9997
Configured but inactive forwards:
        10.180.243.18:9997

┌──(promise㉿kali)-[/opt/splunkforwarder/bin]
└─$ 
```

4. Security Incident Detection

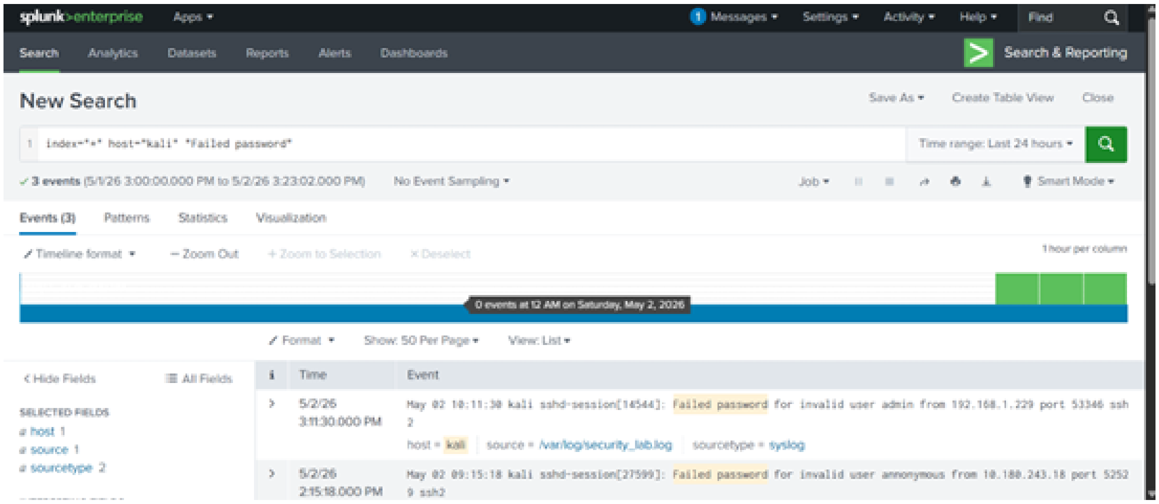
An SSH 'attack' was simulated by attempting unauthorized access to the Kali machine. These events were piped into a dedicated log file and instantly forwarded to the SIEM.

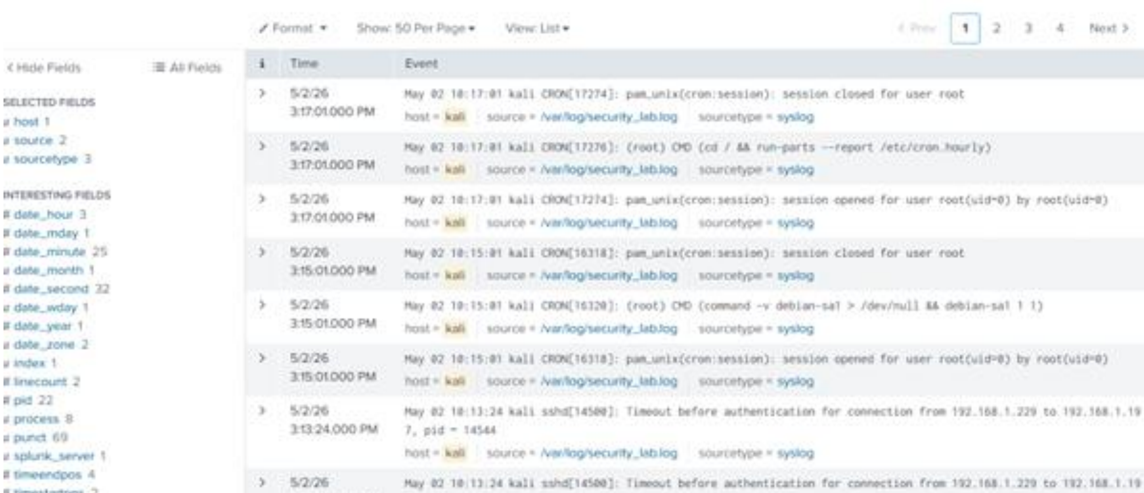
EVIDENCE 04: SIMULATED SSH BRUTE FORCE



```
C:\Users\admin>ssh admin@192.168.1.197
admin@192.168.1.197's password:
Permission denied, please try again.
admin@192.168.1.197's password:
Permission denied, please try again.
admin@192.168.1.197's password:
admin@192.168.1.197: Permission denied (publickey,password).
```

EVIDENCE 05: SIEM ALERT ANALYSIS





5. Analysis: The Timezone Discrepancy

Observation: As seen in Evidence 05, the log text shows 10:11 AM while the Splunk indexer reports 3:11 PM.

The Technical Reason: This is a 5-hour Timezone Offset. The Kali sensor is configured to UTC (Coordinated Universal Time) or a different local zone, while the Windows SIEM instance is recording events based on its local system clock.

Professional Remediation: In an enterprise SOC, we implement NTP (Network Time Protocol) synchronization across all systems to ensure chronologically accurate incident timelines for forensic investigations.